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I. Abstract

This is an individual research paper focused on the impact of MGNREGA on the schooling progression of rural children in India. Two waves of the Indian Human Development Survey (IHDS 2004-5) and (IHDS 2011-12) are used to construct a pooled micro-level dataset of rural children aged 6-14, and grade-for-age is used as a dependent variable in an empirical DiD framework to determine schooling progression. The empirical DiD model is estimated by comparing changes in grade-for-age between the treated (MGNREGA villages) and control (non-MGNREGA villages) groups, controlling for a number of explanatory variables. The results indicate that living in a MGNREGA village increases the probability of being in the correct grade for age by 6-7 percentage points compared to similar non-MGNREGA villages. A robustness check using “grade gap” as the independent variable shows that the results remain consistent and that children in treated villages are comparatively, quarter of a grade closer to being in their expected grade. Gender-interaction results report very modest differences with slightly larger improvements for girls. Overall, MGNREGA is observed to have a positive impact on schooling progression of rural children.

II. Introduction -

Being the world's 5th largest and fastest growing economy, India has been striving towards economic growth and development for several years. Numerous policies have been devised and implemented over the past couple of decades specifically to this effect. One major point of focus has been the improvement in the level of education for rural children. Education is a major determinant of human capital formation and therefore is essential for sustained long-term economic growth. However, even after decades of implementing new and dynamic policy measures, concerns remain regarding children being in the correct grade according to their age and how this has an indirect effect on them with time. In India, most rural children do enroll in primary school but due to various reasons their progression is irregular which in most cases makes dropping out more prevalent.

As a result, the identification of specific policies which enable children to remain in the correct grade for age is important in the context of human capital formation and therefore the broader objective of economic growth and development.

The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) was passed in 2005 and is the policy under consideration for this paper. However, full-scale implementation started in early 2006 and consequently rolled out over several years. MGNREGA promises at least 100 days of employment for every rural household for a specific wage. Any rural household that signs up for unskilled manual work is eligible and receives benefits including but not limited to wage payment within 15 days' work period and an unemployment allowance if work was not provided to them within 15 days of demand. The objective of MGNREGA was to enhance the financial security of rural households and improve village infrastructure by the development of roads, canals, and medical facilities essential for elevating the standard of people in rural areas. MGNREGA therefore has a two-pronged approach, first to provide unemployed members of rural households with work and second, the

completion of work provided in turn would socially and economically elevate these rural people, their families and villages overall.

In this paper, I deal with the following question: Does the Mahatma Gandhi National Rural Employee Guarantee Act (MGNREGA) improve children's grade-for-age progression in rural India?

What I will be testing specifically is that whether being part of a MGNREGA village increases the probability of children being in the right grade for age compared to other children from similar villages with no MGNREGA participation, and how their relationship changes between pre MGNREGA participation (2004-5) and post MGNREGA participation (2011-12). The objective of this paper is to develop a specific but policy relevant outcome, i.e. to generate an indicator representing correct grade for age capturing late entry and repetition altogether.

The data I will be using to tackle this question is from the Indian Human Development Survey (IHDS) which covers a multitude of topics including data on Indian household and children. In this paper, I have focused on rural children between the ages of 6-14 and corresponding data on age, grade in school, household characteristics, and village level MGNREGA participation. I have adopted a difference in differences (DiD) strategy to compare the difference in probability of a rural child being in the expected grade for age between villages that later go on to participate in MGNREGA and villages that do not.

While existing literature on MGNREGA includes its impact on wages, poverty and labor market outcomes, this paper adds to it by trying to establish MGNREGA's impact on schooling progression, using a nationally representative dataset and a transparent empirical DiD approach. Furthermore, by specifically focusing on grade-for-age rather than enrollment alone, and examining potential gender differences in the estimated effects, the objective is to find whether MGNREGA can be truly linked to children's overall educational improvement rather than simply an increase in their presence in school.

This paper is structured as follows. Section II contains the review of related literature such as works on public programmes and child schooling. Section III gives a description of the economic framework and the empirical DiD approach applied. Section IV goes on to describe the handling/cleaning of the IHDS data, and the construction of key variables used for the analysis. Section V presents the relevant empirical results and discusses a gender heterogeneity extension to my original research question. Section VI concludes the paper with a brief discussion about potential limitations and policy implications.

III. Motivation and Literature Review

As mentioned in Section I, India has successfully attained nearly universal primary school enrollment. However, concerns remain regarding the progression of children through the schooling system. In India, many rural children enroll late with respect to their age and before the Right to Education Act was introduced in 2009, repetition was prevalent. As a result, simply enrolling in primary schooling is not enough but tracking children's grade progression is of utmost importance because it has been observed that sluggish progression and over-age students are at a higher risk of dropping out. In this context, grade-for-age has been

predominantly used in education literature as a measure of overall educational progress, and that is precisely the objective of this paper, i.e. to use grade-for-age as an indicator to determine the impact of MGNREGA on the education of rural children.

By providing income and employment definitively to rural households, MGNREGA may have a positive effect on the education of rural children and more specifically on the probability of them being in the right grade for age. This is because, children will not have to supplement parent labor income any longer (which relates to the issue of child labor in India) and will also need to contribute less to agricultural work and resource collection (such as fetching water and collecting firewood). On the other hand, if household employment opportunities rise, this may push older children to substitute into household work like taking care of their siblings, cooking and so on.

As a result, the effect of MGNREGA on the education of children is ambiguous and needs to be empirically tested.

The existing literature on MGNREGA is rich and vast, spanning over several socio-economic questions and issues. Studies such as Bhuvania et al. (2017), “Public Work Programs and Social Capital: An Exploration of MGNREGA in India” discuss about the social and institutional impact of MGNREGA in India while Narayanan, Naraparaju and Gerber (2019), “Synergies in Social Protection: Impacts of India’s MGNREGA and Public Distribution System on the Health and Nutrition of Women and Children”, investigates how MGNREGA interacts with other policies thereby impacting health and nutrition outcomes. Even though schooling is not their primary focus, these papers gather evidence to conclude that MGNREGA impacts and changes the economic and social environment facing households.

A small number of studies deal directly with child level outcomes such as Desai, Joshi and Vanneman (2016), who in their Population Association of America paper “Do Public Works Programs Improve Child Education: Evidence from India”, use IHDS data to study the relationship between public works programmes and measures of child education. This paper provides evidence that access to public work programmes such as MGNREGA can improve children’s educational outcomes. This paper is closely related to my research question in the sense that they also try to establish the impact of MGNREGA on the education of rural children using a set of education indicators and programme exposure but not the grade-for-age indicator used here. Studies on child labor, time use, and health in the context of MGNREGA have provided sufficient evidence that its impacts can be heterogenous across groups.

A substantial amount of international literature exists which explores the impact of cash transfers, conditional cash transfers and public works programmes on child schooling and education. A common theme in most such studies is that the relaxation of credit and income constraints on households tends to support child schooling. However, an increase in employment opportunities and income thereby raises the shadow price of children’s time at home and renders the combined effect ambiguous. In developing countries, several empirical studies have been conducted to analyze the change in grade-for-age as an outcome, arguing that such an indicator is better suited and more closely linked to completion and dropout than basic enrolment rates.

IV. Economic Model and Empirical Approach

The discussion in the previous section suggests that MGNREGA may affect children's schooling in two major ways, one by relaxing liquidity and credit constraints on parents by guaranteeing wage employment and second, due to the increase in time spent on paid work by parents, the demand for a child's time in domestic work such as caring for siblings may rise. This is essentially an optimization problem wherein parents must choose how to allocate a child's time subject to budget and time constraints. MGNREGA shifts this optimization problem by raising income and potentially changing the opportunity cost of children's time.

Children's grade for age is the outcome of interest for the empirical DiD model. For each child i from village v and survey year t , I observe their completed grade, $\text{grade}_{i\text{vt}}$ and age in years, $\text{age}_{i\text{vt}}$. I then construct the expected grade assuming entry into class 1 at age 6 as -

$$\text{expected_grade}_{i\text{vt}} = \text{age}_{i\text{vt}} - 5,$$

And define the binary indicator

$\text{ontrack}_{i\text{vt}} = 1(\text{grade}_{i\text{vt}} \geq \text{expected_grade}_{i\text{vt}})$, which equals 1 if the child is at least in the expected grade for their age and 0 otherwise. Both late entry and repetition are captured in this variable. As a robustness check, I use a continuous "grade gap" instead of $\text{ontrack}_{i\text{vt}}$ as an outcome, and define it as follows -

$\text{grade}_{i\text{vt}} - \text{expected_grade}_{i\text{vt}}$, which measures by how many grades a child is ahead/behind of the expected grade for age.

Using the 2011-12 IHDS data, I define treatment at the village level. A village is defined as treated if at least one household from it reported work or income from MGNREGA and control if no such work or income was reported by any household from that village. treat_v is a treatment dummy which equals 1 for treatment and 0 for control. This treatment status is then attached to all children living in those villages in both waves. For observations from 2011-12 and 2004-05 a time dummy post_t equals 1 and 0 respectively. Since the 2004-05 wave refers to a period before MGNREGA was implemented, all villages are effectively untreated at $v=0$ even if they classified under treated based on their future participation.

The empirical DiD is calculated for the following linear probability model -

$$\text{ontrack}_{i\text{vt}} = \alpha + \beta_1 \text{treat}_v + \beta_2 \text{post}_t + \delta(\text{treat}_v \times \text{post}_t) + X_{i\text{vt}} \cdot \gamma + \mu_s + \varepsilon_{i\text{vt}}$$

Where $X_{i\text{vt}}$ is a vector of child and household level controls, μ_s are state fixed effects and $\varepsilon_{i\text{vt}}$ is the error term. The control vector $X_{i\text{vt}}$ consists of variables that are likely to be correlated with MGNREGA treatment and schooling outcomes. I control child age in years and its square to enable a non-linear relationship between age and the probability for being on track. A female dummy, per capita household consumption and highest adult education in a household are also used as controls and these altogether constitute the $X_{i\text{vt}}$ vector. Time-invariant differences between states with respect to schooling systems, infrastructure and governance are absorbed

by the state fixed effects. Standard errors are clustered at the village level which is the level at which treatment has been defined.

δ is the difference-in-differences estimate and therefore the parameter of interest in this analysis. It intuitively refers to by how many percentage points did the probability of a child being on track increase/decrease over time in MGNREGA villages compared to other similar non-MGNREGA villages, after controlling for the specified controls and state fixed effects. One major assumption of this model is the parallel trends assumption. In this context, the parallel trends assumption refers to the fact that in the absence of MGNREGA, treated and control villages would have seen similar changes in grade-for-age conditioned on the included controls. δ can therefore also be interpreted as the causal impact of living in a MGNREGA village on grade progression.

To analyze heterogeneity, I extended the baseline DiD specification to a triple-interaction model- $\text{ontrack}_{ivt} = \alpha + \beta_1 \text{treat}_v + \beta_2 \text{post}_t + \beta_3 \text{girl}_{ivt} + \delta_1(\text{treat}_v \times \text{post}_t) + \delta_2(\text{treat}_v \times \text{post}_t \times \text{girl}_{ivt}) + X_{ivt} \cdot \gamma + \mu_s + \varepsilon_{ivt}$

Where girl_{ivt} is a dummy equaling 1 for girls and 0 otherwise. In this model, delta 1 captures the DiD effect for boys while delta 2 captures the extra effect for girls. As a result, the total DiD effect on girls is captured by delta 1 + delta 2. Finally, the entire model is re-estimated using the same specifications but using gradedgap_{ivt} as the dependent variable to ensure that my findings do not depend on the way I turn grade-for-age into a binary indicator.

V. Data

As mentioned in Section I, the IHDS data jointly collected by the University of Maryland (UMD) and the National Council of Applied Economic Research (NCAER) has been used for this analysis. The IHDS was conducted as a nationally representative survey for two waves (2004-05 and 2011-12) and contains detailed information on household characteristics, village context, and children's schooling and education. As a result, IHDS is uniquely suited to undertake a before and after analysis in the context of MGNREGA.

In each wave I retain rural households from the household files and construct per capita monthly consumption expenditure (COPM) and the highest adult completed years of schooling in the household (). I use the 2011-12 household file to identify whether any household member reports income or work hours from MGNREGA. I then use this information at the village level by individually looking at each village, counting the number of households surveyed and how many of them reported MGNREGA participation. Then, as mentioned in section IV the treat_v variable is created.

Child level information is obtained from the individual files and in each wave, children are merged to their household records with the help of the household ID's. I restrict the sample by only considering rural children aged 6-14, keeping those with non-missing information on age, current grade of schooling, and key controls. Age is measured in completed years and in each wave, grade is recorded separately (ED5 in 2004-05 and ED6 in 2011-12). Negative or special

values in the grade variables, I record as missing and then construct the $\text{expected_grade}_{ivt}$ variable and ontrack_{ivt} , the indicator variable as per section IV. To enable the usage of same notation across waves, a generic grade variable is defined and the grade gap outcome is derived to perform the robustness check.

The final pooled samples consist of 28,661 rural children between the ages of 6 and 14 in 2004-05 and 25,439 in 2011-12 living in 1,417 rural villages across the Indian states. Their summary statistics reveal that both treatment and control villages are similar to each other in the context of age and gender composition. In both groups, average age of child is approximately 10 years and about 48 per cent of total children are female. However, as will be later explored in Section VI, it can be easily observed from the summary statistics that villages which later adopt MGNREGA start out poorer and with a lower level of adult education than villages which do not adopt MGNREGA.

Since MGNREGA explicitly targets rural areas which are disadvantaged as compared to other rural areas, this inconsistency in levels is to be expected rather than problematic. The empirical strategy, therefore, centers on a DiD design in changes, by controlling household consumption expenditure per capita, adult education and state fixed effects in all regressions, rather than levels. All analyses are unweighted and have standard errors clustered at the village level, i.e. the level at which treatment is defined.

VI. Empirical Results

Table 1- Balance Table for 2004-2005

Variable	Control Villages ($\text{treat_v} = 0$)	Treated Villages ($\text{treat_v} = 1$)
Age (years)	10.05	9.95
Girl (Dummy)	0.476	0.480
ontrack	0.305	0.228
COPC	681.84	585.01
Highest Adult Education (years)	6.37	5.43
Observations	8,981	19,680

Table 2- Mean on-track rates by treatment status and period

Group	2004-05 (post=0)	2011-12 (post=1)
Control Villages ($\text{treat_v} = 0$)	0.305	0.340
Treated Villages ($\text{treat_v} = 1$)	0.228	0.310

Table 1 is suggestive of the fact that treated villages start off as poorer and have lower levels of adult education, while child age and gender distributions are very similar across the

treatment and control groups. This confirms the need to control for household consumption expenditure and adult education while estimating the main DiD effect.

From Table 2 it is observed that children in MGNREGA villages experience a higher improvement in grade for age compared to those in non-MGNREGA villages. It can be observed that while in 2004-05 only about 22 percent of children in MGNREGA villages were on track for age by 2011-12, 31 percent of them were on track for age. Comparatively, children in non-MGNREGA villages started from a higher percentage being in the right grade for age in 2004-05 (30 percent), and this number increased to 34 percent by 2011-12. Therefore, it can be roughly stated that both groups improve over time, MGNREGA villages by 8 percentage points and non-MGNREGA villages by 3½ percentage points, implying a raw DiD of around 4.7 percentage points. However, whether this holds while controlling for observable characteristics and state level differences must be checked.

Baseline DiD estimates -

Table 3 describes the main DiD regressions (baseline DiD and DiD with gender as a control), where the outcome is the indicator for being on track for age. Age, age squared, per-capita consumption (COPC), highest adult education and state fixed effects are included in both specifications, and the standard errors are clustered at the village level.

Table 3 -

	(1) On track	(2) On track + girl
MGNREGA village (treat v)	-0.037	-0.037
post (2011-12)	0.020	0.020
treat x post	0.063* (0.014) **	0.063* (0.014) **
girl	-	0.012 *** (0.014)
age, age ²	yes	yes
COPC, highest adult education	yes	yes
State fixed effects	yes	yes
Observations	54,058	54,058
R ²	0.0998	0.1

Notes: Column 1 contains the on-track indicator while column 2 contains the on-track indicator with the girl dummy.

Legend: *** p<0.01, ** p<0.05 and *p<0.10

Patterns highlighted by Table 3-

- The treat_v coefficient is negative and statistically significant. Consistent with the descriptive differences in table 1, this means that children in MGNREGA villages were less likely to be on track for age in the pre-period (2004-05) even after controlling COPC, highest adult education, age and state fixed effects.

- The post coefficient is marginally significant, describing a modest improvement in on-track rates between 2004-05 and 2011-12.
- The key parameter treat x post is highly significant ($p < 0.01$) and is around 0.063 for both specifications. Therefore, the probability of being on track for age increased by about 6-7 percentage points more in MGNREGA village compared to otherwise similar non-MGNREGA villages.
- Adding the gender dummy does not significantly change the DiD coefficient. The girl coefficient is positive and significant which implies that after controlling other factors, girls are slightly more likely to be on track for age compared to boys.

Gender Heterogeneity and Robustness check -

Table 4 -

	(1) On track (triple DiD)	(2) Grade Gap (years behind age)
MGNREGA village (treat_v)	-0.029** (0.012)	-0.119** (0.047)
post (2011-12)	0.012 (0.014)	0.196*** (0.044)
treat x post	0.052* (0.016)	0.251* (0.050)
girl	0.008 (0.010)	-0.049* (0.017)
treat x girl	-0.016 (0.011)	-
post x girl	0.017 (0.013)	-
treat x post x girl	0.022 (0.016)	-
Controls and state FE	yes	yes
Observations	yes	54,058
R ²	0.1003	0.2508

Notes: Column (1) contains the triple interaction specification for “on track”. Column 2 has the continuous “grade gap” (expected grade – actual grade) as the outcome. *** $p < 0.01$, ** $p < 0.05$ and * $p < 0.10$.

In the triple-interaction model, treat_v x post is the DiD term for boys and it remains positive and significant at about 0.052, implying that boys in treated villages are 5 percentage points more likely to be on track for age compared to those in control villages. The triple interaction coefficient is positive but not statistically significant at conventional levels. However, when we combine the interaction terms to find the total DiD- effect on girls, we see that it is about 7.4 (5.2+2.2) percentage points for girls which is slightly higher.

Column 2 contains the continuous grade-gap outcome defined as expected grade minus actual grade. The coefficient of treat x post is 0.251 which implies that children in MGNREGA villages end up around a quarter of a class closer to where they are supposed to be in

comparison to similar children in non-MGNREGA villages. This result is consistent with the main DiD and indicates that MGNREGA can be associated not only with a higher probability of being on track but also with a smaller shortfall in completed grades.

VII. Brief Conclusion

The analysis in this paper shows that MGNREGA can be meaningfully assumed to be associated with children's schooling progression. The results indicate that, between 2004-05 and 2011-12 children in MGNREGA villages observed a 6-7 percentage points of improvement in the probability of being in the correct grade for age. The robustness checks using the continuous grade gap show that children in treated villages are about a quarter of a grade closer to their expected grade. In the context of gender interaction estimates it can be concluded that while there are slightly higher gains for girls, these differences are modest.

All findings are consistent with the belief that public employment programmes can ease household constraints and therefore improve children's schooling despite not being specifically targeted towards education. The analysis is focused on only two major survey waves and cannot therefore directly test the parallel-trends assumption. It also cannot fully capture unobserved differences between treated and control villages. Exploiting more frequent data and additional reforms in the examination of dynamic effects could be plausible for future work, for example utilizing study times or test scores.

References

Desai, S. and Vanneman, R. (2010) *India Human Development Survey (IHDS), 2005* [dataset]. Ann Arbor, MI: Inter-university Consortium for Political and Social Research (ICPSR). <https://doi.org/10.3886/ICPSR22626.v12> (Accessed: 8 December 2025).

Desai, S. and Vanneman, R. (2018) *India Human Development Survey-II (IHDS-II), 2011–12* [dataset]. New Delhi and College Park, MD: National Council of Applied Economic Research (NCAER) and University of Maryland. Available at: <https://doi.org/10.3886/ICPSR36151.v6> (Accessed: 8 December 2025).

Government of India (2005) *The National Rural Employment Guarantee Act, 2005*. New Delhi: Ministry of Law and Justice. Available at: <https://legislative.gov.in> (Accessed: 8 December 2025).

Government of India (2009) *The Right of Children to Free and Compulsory Education Act, 2009*. New Delhi: Ministry of Law and Justice. Available at: <https://legislative.gov.in>

Kumar, P. and Chakraborty, D. (2016) *MGNREGA: Employment, Wages and Migration in Rural India*. Routledge.